

QUANTITATIVE FINANCE
PART II - DECISION THEORY AND PORTFOLIO
CHOICE

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A Very Brief History of Decision Theory

Decision theory is the theory of **rational** decision making. The history of decision theory can be divided into three distinct phases:

- 1 the Old period;
- 2 the Pioneering period;
- 3 the Axiomatic period.

The Old Period

The ancient Greeks were familiar with the distinction between “*right*” and “*rational*” acts.

- o A decision is **right** if and only if its actual outcome is at least as good as that of every other possible outcome.
- o A decision is **rational** if and only if the decision maker chooses to do what she has most reason to do at the point in time at which the decision is made.
- o **Decision theorists are primarily concerned with rational decisions rather than right ones.** In many cases it seems impossible to foresee, even in principle, which act is right until the decision has already been made. It seems much more reasonable to claim that it is always possible to foresee whether a decision is rational. This is because theories of rationality operate on information available at the point in time the decision is made rather than on information available at some later point in time.

The Old Period

Herodotus takes “right” to mean rational and “contrary to good counsel” to mean irrational.

“For even if the event turns out contrary to one’s hope, still one’s decision was right, even though fortune has made it of no effect: whereas if a man acts contrary to good counsel, although by luck he gets what he had no right to expect, his decision was not any the less foolish.” (Herodotus VII: 10)

Aristotle proposes what may best be conceived of as an embryo for the logic of rational preference:

“If A be without qualification better than B, then also the best of the members of A is better than the best of the members of B; e.g. if Man be better than Horse, then also the best man is better than the best horse.” (Topics III: ii)

The Old Period

However, the Greeks did not develop a **theory** of rational decision making.

They were not familiar with the **concept of probability**. The Greek language contains expressions for everyday notions of likelihood and chance, but none of the great mathematicians working at that time formalized these ideas into a mathematical calculus.

There is no evidence that Greek thinkers formulated any **precise decision rules**, at least not in the same general and abstract form as modern decision rules.

The Pioneering Period

Fifteen hundred years after the demise of the Greek culture, it began in 1654 as Blaise Pascal and Pierre de Fermat began to exchange letters about probability theory, to work out an **exact, mathematical solution** to the following problem.

“What is the likelihood of getting at least one pair of sixes if a pair of fair dice is thrown 24 times?” (Answer: $1 - (35/36)^{24} \approx 0.49$.)

Another major breakthrough was the 1662 publication of a book known as *Port-Royal Logic*, which contains the first clear formulation of the **principle of maximizing expected value**:

“In order to judge of what we ought to do in order to obtain a good and to avoid an evil, it is necessary to consider not only the good and evil in themselves, but also the probability of their happening and not happening, and to regard geometrically the proportion which all these things have, taken together.” (Arnauld and Nicole 1662: 367)

The Pioneering Period

Given the choice between maintaining the *status quo* and a gamble in which one would either win or lose one million gold ducats with equal probability, that is, a fair game,¹ *Port-Royal Logic* recommends us to be indifferent between the two options. Intuitively, we know that **cannot** be right.

In 1738, Daniel Bernoulli introduced the notion of **moral value** as an improvement on the imprecise terms “good” and “evil”. The moral value of an outcome is a technical term referring to how good or bad that outcome is from the decision maker’s point of view. A multimillionaire might be indifferent between the status quo and the above fair gamble, simply because he is already very well off. For the rest of us, this is unlikely to be the case: the negative moral value of losing a million is for most of us much greater than the positive moral value of winning a million.

In the modern literature, the term “moral value” has been replaced by the notion of **utility**.

¹We will study it carefully soon.

The Axiomatic Period

Modern decision theory is dominated by attempts to axiomatize the principles of rational decision making, which explains why we call it the Axiomatic period.

- 1 Frank Ramsey, in his paper “*Truth and Probability*”, written in 1926 but published posthumously in 1931, proposed a set of eight axioms for how rational decision makers ought to choose among uncertain prospects. He pointed out that every decision maker behaving in accordance with these axioms will act in a way that is **compatible** with the principle of maximizing expected value, by implicitly assigning numerical probabilities and values to outcomes. However, it does **not** follow that the decision maker’s choices were **actually** triggered by these implicit probabilities and utilities.
- 2 von Neumann and Morgenstern, in the second edition of their book “*Theory of Games and Economic Behavior*” in 1947,² presented a set of axioms for how rational decision makers ought to choose among lotteries. They showed that every decision maker behaving in accordance with their axioms implicitly behaves in accordance with the principle of **maximizing expected utility**, and implicitly assigns **numerical utilities** to outcomes.

²The first edition of the book was published in 1944, with the first chapter devoted to individual decision making under risk, whereas they are complained for not giving any technically precise definition of the notion of utility.

The Axiomatic Period

- 3 The 1950s turned out to be a golden age for decision theory. Leonard Savage, in his 1954 book “*The Foundations of Statistics*”, presented yet another influential axiomatic analysis of the principle of maximizing expected utility.

The above discussion is based on Peterson, Martin, 2017. *An Introduction to Decision Theory*, 2nd edition, Cambridge University Press.

Investment Decision over a Single Period



Your future wealth will depend on the **random** realization of the asset prices.

The Expected Utility framework provides the weapon for decision-making under uncertainty, that is, **ranking the random payoffs**.

It trades off expected returns with volatility, as we will show later, and more things like skewness and kurtosis.

Utility Functions

A utility function $U(x)$ is defined on real numbers and gives a real utility score. You compare two random outcomes by comparing the expected utility $E[U(\tilde{x})]$ of the two.

John von Neumann and Oskar Morgenstern pioneered the expected utility framework in 1944 and 1947, and they showed that preferences over random investment opportunities can be represented by a certain expected utility function when preferences satisfy some axiomatic conditions.

One general restriction on the utility function is that it is **strictly increasing** (if $x > y$, then $U(x) > U(y)$) and **continuous** (to preclude dictionary/alphabetical order). Other than this, utility function could in theory **take any form**. And it could **vary** among individual investors.

Commonly Used Utility Functions

1 Exponential:

$$U(x) = -e^{-ax},$$

for some $a > 0$.

2 Power:

$$U(x) = \frac{x^{1-\gamma}}{1-\gamma},$$

for some $\gamma > 0$. Writing $U(x) = \frac{x^{1-\gamma}-1}{1-\gamma}$ and letting $\gamma \rightarrow 1$, we obtain the **logarithmic** utility $U(x) = \ln(x)$.

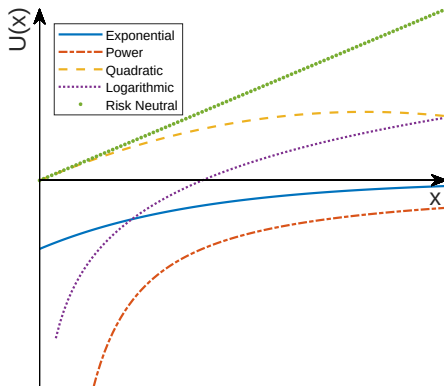
3 Quadratic:

$$U(x) = x - bx^2,$$

for some $b > 0$. We need $x < 1/(2b)$ for it to be increasing.

4 When $\gamma \rightarrow 0$ or $b \rightarrow 0$, we obtain the **risk-neutral** utility $U(x) = x$ and hence $E[U(\tilde{x})] = E[\tilde{x}]$, wherein investors only care about expected payoffs.

Commonly Used Utility Functions



Decision under Uncertainty: An Illustration

Suppose an investor has a logarithmic utility $U(x) = \ln(x)$. She would invest either in a riskfree bond which generates a future payoff of \$100, or in a risky stock which produces a payoff of \$80 and \$120 with corresponding probabilities 50% and 50%.

Her expected utility of investing in the bond is

$E[U(\tilde{x})] = \ln(100) = 4.6052$, while her expected utility of investing in the stock is

$E[U(\tilde{x})] = 0.5 \ln(80) + 0.5 \ln(120) = 4.5848 < 4.6052$.

Therefore, she would take the bond.

Pros for Utility Functions

- 1 **Merits of simplicity:** Just compare the expected values of the utility scores to make decisions.
- 2 **Flexibility of various functional forms:** There are not many restrictions on the functional forms (we will discuss choosing a specific utility function soon).
- 3 **Strong theoretic justifications:** They can be derived from a small set of axioms describing the preferences of rational investors.

LLN v.s. Expected Utility

Decision theorists have proposed two fundamentally different arguments for the expected utility principle.

- The first argument is based on **the law of large numbers**; it seeks to show that in the long run you will be better off if you maximize expected utility.
- The second argument aims at deriving **the expected utility principle** from some more fundamental axioms for rational decision making, which make no reference to what will happen in the long run.

LLN: An Example

You are offered 1 unit of utility for sure or a lottery ticket that will yield either 10 units with a probability of 0.2, or nothing with a probability of 0.8. The expected utility of choosing the lottery ticket is

$$2 = 0.2 * 10 + 0.8 * 0 > 1.$$

According to the law of large numbers, you **cannot** be sure that you will actually be better off if you choose the lottery, given that the choice is offered to you only once.

However, what you do know for sure is that if you face the same decision over and over again, then the probability that you will not be better off by choosing the lottery can be made **arbitrarily small** by repeating the same decision over and over again.

To be precise, if you repeat the decision in question infinitely many times, then **the probability that the average utility and the expected utility differ by more than an arbitrarily small units decreases to zero.**³

³To be precise, invoke the Weak Law of Large Numbers.

Three Worries regarding the LLN Interpretation

- 1 Keynes famously objected to the law of large numbers that, “in the long run we are all dead”. (Keynes, John. M., 1923. *A Treatise on Probability*, Macmillan&Co.)
- 2 Another mathematical theorem, known as *gambler's ruin*, partially defeats LLN. Imagine that you and I flip a fair coin, and that I pay you \$1 every time it lands heads up, and you pay me the same amount when it lands tails up. If we toss the coin sufficiently many times, each player will at some point encounter a sequence of heads or tails that is longer than he can afford and go bankrupt, given that each player starts with pots containing finite amounts of money.
- 3 Many decisions under risk are unique, made only once, such as a decision to start a war, or appointing a chief executive, or electing a new president.

Equivalent Utility Functions

If $U(x)$ and $V(x)$ **always** give the same ranking of **all** random payoffs, they are said to be equivalent. It can be proved that, the necessary and sufficient condition for equivalency is that:

$$V(x) = aU(x) + b, \quad (1)$$

where a and b are arbitrary constants with $a > 0$. Hence expected utility functions are equivalent up to **linear monotonic transformations**. They are “**cardinal**” instead of “ordinal”.

This enables us to write utility functions in **convenient** forms. For instance, we could write the power utility $U(x) = \frac{x^{1-\gamma}-1}{1-\gamma}$, $\gamma > 0$ as $U(x) = x^\gamma$, $\gamma < 1$ (Both an equivalent transformation and a parameter redefinition are applied.). But in this case, we would not easily see that the logarithmic utility is a special case of the power.

However, we could solve the problem by writing it as $U(x) = x^\gamma/\gamma$ and letting $\gamma \rightarrow 0$.

We will talk about another reason for further rewriting in the following. It turns out that all the forms we have mentioned are **useful under certain circumstances**.

Concave Utility and Risk Aversion

- Recall our discussion of convexity in Part II. A utility function on $[\underline{x}, \bar{x}]$ is **convex** if for any admissible x_1, x_2 and $0 \leq \alpha \leq 1$,

$$U(\alpha x_1 + (1 - \alpha)x_2) \leq \alpha U(x_1) + (1 - \alpha)U(x_2). \quad (2)$$

An investor is risk seeking/loving if her utility function is convex.

- A utility function on $[\underline{x}, \bar{x}]$ is **concave** if for any admissible x_1, x_2 and $0 \leq \alpha \leq 1$,

$$U(\alpha x_1 + (1 - \alpha)x_2) \geq \alpha U(x_1) + (1 - \alpha)U(x_2). \quad (3)$$

An investor is risk averse if her utility function is concave.

- A utility function on $[\underline{x}, \bar{x}]$ is **linear** if for any admissible x_1, x_2 and $0 \leq \alpha \leq 1$,

$$U(\alpha x_1 + (1 - \alpha)x_2) = \alpha U(x_1) + (1 - \alpha)U(x_2). \quad (4)$$

An investor is risk neutral if her utility function is linear.

Concave utility and Fair Games

If an investor is **always** unwilling to accept **any** fair game (asset) with an expected payoff of zero, her utility function is concave.

To see this, suppose an individual has current wealth W and utility function U . She is offered a fair game with payoff $\tilde{x} = x_1$ with probability p and payoff $\tilde{x} = x_2$ with probability $1 - p$. By definition, $E[\tilde{x}] = px_1 + (1 - p)x_2 = 0$.

The individual's unwillingness to accept any fair game means, for arbitrary W , x_1 , and x_2 ,

$$U(W) \geq E[U(W + \tilde{x})] = pU(W + x_1) + (1 - p)U(W + x_2).$$

Note that

$$U(W) = U(W + px_1 + (1 - p)x_2) = U(p(W + x_1) + (1 - p)(W + x_2)).$$

Hence

$$U(p(W + x_1) + (1 - p)(W + x_2)) \geq pU(W + x_1) + (1 - p)U(W + x_2),$$

which is just the definition of the concavity of U .

Concave utility and Fair Games

Conversely, concavity of utility functions also means unwillingness to accept fair games.

Simply apply Jensen's inequality⁴ from statistics. For a concave function U and a random payoff $\tilde{\omega}$,

$$E[U(\tilde{\omega})] \leq U(E[\tilde{\omega}]).$$

Substituting $\tilde{\omega} = W + \tilde{x}$, we get

$$E[U(W + \tilde{x})] \leq U(E[W + \tilde{x}]) = U(W).$$

⁴Specifically, suppose f is a probability density function on $\mathcal{D} \in \mathbb{R}$. Hence, f is nonnegative and $\int_{\mathcal{D}} f(x)dx = 1$. If g is concave on $\mathcal{D}$, then

$$\int_{\mathcal{D}} g(x)f(x)dx \leq g\left(\int_{\mathcal{D}} xf(x)dx\right).$$

Risk Aversion

A utility function is strictly increasing if $U'(x) > 0$. It is strictly concave if $U''(x) < 0$. The stronger the concavity, or the bend of the function, the higher the risk aversion.

We can formally measure risk aversion using the **Arrow-Pratt absolute risk aversion coefficient**:

$$A(x) = -\frac{U''(x)}{U'(x)}. \quad (5)$$

The negative sign makes the coefficient **positive** for a risk averse utility — the larger the bend, the higher the coefficient.

The denominator normalizes the coefficient such that an equivalent linear transformation does **not** change the coefficient.

For the exponential utility,

$$U(x) = -e^{-ax}, \quad \Rightarrow \quad A(x) = a.$$

And for the power utility,

$$U(x) = \frac{x^{1-\gamma}}{1-\gamma}, \quad \Rightarrow \quad A(x) = \gamma/x.^5$$

The power utility exhibits lower risk aversion for higher wealth, capturing the fact that people become less risk averse and willing to take more risk as their wealth grows.

Still for the quadratic utility,

$$U(x) = x - bx^2, \quad \Rightarrow \quad A(x) = 2b/(1 - 2bx).$$

For $b > 0$ and $x < 1/(2b)$, risk aversion increases with wealth.

⁵Clearly, writing $U(x)$ in this form delivers a simple expression and an intuitive interpretation of γ .

Certainty Equivalent

We have established the link between risk aversion and concavity of utility functions. How to measure risk aversion in an economic sense? An answer is through **risk premium**, which is defined as the difference between the expected value and certainty equivalent of a random payoff (John W. Pratt, 1964).

The certainty equivalent of a random variable $\tilde{x}$ is defined as the certain (riskfree) payoff C that achieves the same utility level. That is,

$$U(C) = E[U(C)] = E[U(\tilde{x})].$$

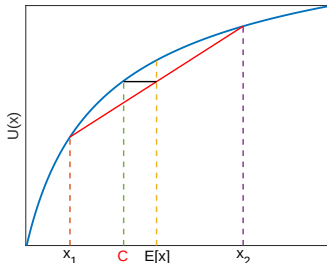
Note that C corresponding to $\tilde{x}$ is the same for all equivalent utility functions.

Define the risk premium $\pi = E[\tilde{x}] - C$. Then

$$U(E[x] - \pi) = E[U(\tilde{x})].$$

Hence π is **the amount the individual is willing to pay to avoid risk**.

Certainty Equivalent and Risk Premium



For a risk averse investor, if $\tilde{x}$ pays x_1 and x_2 with probabilities α and $1 - \alpha$, respectively,

$$\begin{aligned}
 U(C) &= E[U(\tilde{x})] = \alpha U(x_1) + (1 - \alpha)U(x_2) \\
 &\leq U(\alpha x_1 + (1 - \alpha)x_2) = U(E[\tilde{x}]), \text{ (Concavity)} \\
 \implies C &\leq E[\tilde{x}] \text{ and } \pi \geq 0, \text{ as } U(x) \text{ is strictly increasing.}
 \end{aligned}$$

Decision through Certainty Equivalent

As a utility function is increasing, an investor would prefer a payoff with a higher certainty equivalent value. This provides an alternative to comparing expected utility values.

Suppose an investor has a logarithmic utility $U(x) = \ln(x)$. She would invest either in a riskfree bond which generates a future payoff of \$100, or in a risky stock which produces a payoff of \$80 and \$120 with corresponding probabilities 50% and 50%.

Her expected utility of investing in the stock is $E[U(\tilde{x})] = 0.5 \ln(80) + 0.5 \ln(120) = 4.5848$, with an according certainty equivalent of $C = \exp(E[U(\tilde{x})]) = 97.98 < 100$.

Therefore, she would take the bond.

Specification of Utility Functions

There are various systematic procedures to assigning utility functions to investors. Here we discuss two relatively more “quantitative” methods.

(1) Direct Measurement through Certainty Equivalent Value

Select two total⁶ wealth levels A and B .

Now I offer you a lottery that awards you A with probability p and B with probability $1 - p$. Hence your expected payoff is $e_p = pA + (1 - p)B$.

How much certain wealth would you pay to participate in this lottery?

⁶Not wealth increments from, e.g., entering into a bet. The prospect theory, developed by Daniel Kahneman and Amos Tversky in 1979, builds on the Prospect Utility which derives felicity from wealth increments instead of total levels. It challenges the expected utility theory and earned Daniel Kahneman the Nobel Memorial Prize in Economics in 2002.

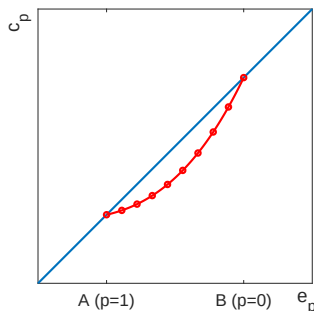
Specification of Utility Functions

(1) Direct Measurement through Certainty Equivalent Value

Say you would pay c_p for an expected payoff of $e_p = pA + (1 - p)B$.

If you are risk averse, then $c_p < e_p$, $0 < p < 1$.

The figure for the function $c(e)$:



Specification of Utility Functions

(1) Direct Measurement through Certainty Equivalent Value

For any utility function $V(x)$, we know that we can represent it in an equivalent form as $U(x) = aV(x) + b, a > 0$. We can choose a and b such that:

$$U(A) = A, \text{ and } U(B) = B.$$

Then we know that

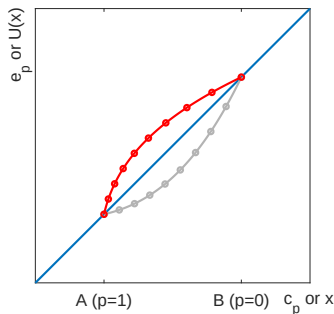
$$E[U(\tilde{x})] = pU(A) + (1 - p)U(B) = pA + (1 - p)B = e_p.$$

That is e_p is the expected utility for the lottery p . Further, e_p is the utility of the certain wealth c_p .

Specification of Utility Functions

(1) Direct Measurement through Certainty Equivalent Value

Take the inverse of $c(e)$ to get $e(c)$, which is just $U(x)$. Flipping the axes we get the figure:



Specification of Utility Functions

(2) Parameterized Families

We assign a parameterized family for the utility function, and what's left is to determine the parameter value.

We use the following three most commonly used families for illustration.

Exponential: $U(x) = -e^{-ax}$, $a > 0$,

Power: $U(x) = \frac{x^{1-\gamma}}{1-\gamma}$, $\gamma > 0$,

Quadratic: $U(x) = x - bx^2$, $b > 0$, $x < 1/(2b)$.

Remember: If your risk tolerance does not change with wealth, choose Exponential; if it increases with wealth, choose Power; if it decreases with wealth, choose Quadratic.

Specification of Utility Functions

(2) Parameterized Families: A Simple Example

We take the Exponential utility $U(x) = -e^{-ax}$, $a > 0$. Since it only has one parameter, one single lottery would be enough to determine a .

Say $A = 1$, $B = 3$, and $p = 0.5$. You think the lottery is equivalent to a certain wealth of $c = 1.75$. Then

$$\begin{aligned} -e^{-1.75a} &= -0.5e^{-a} - 0.5e^{-3a}, \\ \implies a &= 0.52. \end{aligned}$$

You can use Excel Solver to find the value of a . See *CurveFitting.xlsx*.

Specification of Utility Functions

(2) Parameterized Families: An Example of Curve Fitting

We take the Power utility $U(x) = \frac{x^{1-\gamma}}{1-\gamma}$, $\gamma > 0$.

Suppose $A = 1$, $B = 3$, and $\gamma = 3$. We can calculate the certainty equivalent values for $p = 0, 0.2, 0.4, 0.6, 0.8, 1$. We list the results in the following table.

p	0	0.2	0.4	0.6	0.8	1
c_p	3	1.86	1.46	1.25	1.10	1
e_p	3	2.6	2.2	1.8	1.4	1

Now, we ask the opposite question.

We observe the certainty equivalents of several lotteries for an investor with power utility. But we do not know γ . **How to infer it?**

Specification of Utility Functions

(2) Parameterized Families: An Example of Curve Fitting

First, we transform $U(x)$ into an equivalent $V(x) = aU(x) + b$, such that

$$\begin{aligned}
 &V(A) = A \text{ and } V(B) = B, \\
 \implies &a \frac{A^{1-\gamma}}{1-\gamma} + b = A \text{ and } a \frac{B^{1-\gamma}}{1-\gamma} + b = B, \\
 \implies &a = \frac{(1-\gamma)(A-B)}{A^{1-\gamma} - B^{1-\gamma}} \text{ and } b = \frac{BA^{1-\gamma} - AB^{1-\gamma}}{A^{1-\gamma} - B^{1-\gamma}}.
 \end{aligned}$$

Now we need $V(c_p) = e_p$.

Specification of Utility Functions

(2) Parameterized Families: An Example of Curve Fitting

Generally, we find the γ that solves:

$$\min_{\gamma} \sum_p (V(c_p) - e_p)^2, \quad (6)$$

where c_p may be measured with errors. Hence more observed c_p s may be used to improve the estimation of γ , since the random errors in the observed c_p s may (partially) cancel with one another.

See *CurveFitting.xlsx*.

Question 1

HARA Utility Function

The class of HARA utility functions is defined as

$$U(x) = \frac{1-\gamma}{\gamma} \left(\frac{ax}{1-\gamma} + b \right)^{\gamma}, \quad (7)$$

where $x \in \left\{ x \mid \frac{ax}{1-\gamma} + b > 0 \right\}$, and a , b , and γ are such that $U(x)$ is increasing and concave in x . Pick a , b , and γ (and perform equivalent transformations if needed) to obtain the following special cases:

- 1 Linear or risk neutral: $U(x) = x$
- 2 Quadratic: $U(x) = x - 0.5cx^2$
- 3 Exponential: $U(x) = -e^{-ax}$
- 4 Power: $U(x) = cx^{\gamma}$
- 5 Logarithmic: $U(x) = \ln(x)$

Question 1 - Answer

HARA Utility Function

$$U(x) = \frac{1-\gamma}{\gamma} \left(\frac{ax}{1-\gamma} + b \right)^{\gamma}, \quad x \in \left\{ x \mid \frac{ax}{1-\gamma} + b > 0 \right\}.$$

1 Linear or risk neutral: $U(x) = x$

Answer: $a = 1$, $b = 0$, and $\gamma = 1$.

For $a = 1$ and $b = 0$,

$$U(x) = (1-\gamma)^{1-\gamma} x^{\gamma} / \gamma.$$

$$\begin{aligned} \text{Then } \lim_{\gamma \rightarrow 1-} (1-\gamma)^{1-\gamma} &= e^{\lim_{y \rightarrow 0+} \frac{\ln y}{y-1}} \stackrel{\text{L'Hôpital's rule}}{=} e^{\lim_{y \rightarrow 0+} \frac{y^{-1}}{-y^{-2}}} = \\ e^{\lim_{y \rightarrow 0+} -y} &= 1. \end{aligned}$$

Question 1 - Answer

HARA Utility Function

$$U(x) = \frac{1-\gamma}{\gamma} \left(\frac{ax}{1-\gamma} + b \right)^{\gamma}, \quad x \in \left\{ x \mid \frac{ax}{1-\gamma} + b > 0 \right\}.$$

2 Quadratic: $U(x) = x - 0.5cx^2$

Answer: $b = 1/a$ and $\gamma = 2$.

For $\gamma = 2$,

$$U(x) = -0.5(-ax + b)^2 = abx - 0.5a^2x^2 - 0.5b^2.$$

Then $c = a^2$ and $b = 1/a$ give the equivalent

$$U(x) = x - 0.5cx^2 - 0.5b^2.$$

Question 1 - Answer

HARA Utility Function

$$U(x) = \frac{1-\gamma}{\gamma} \left(\frac{ax}{1-\gamma} + b \right)^{\gamma}, \quad x \in \left\{ x \mid \frac{ax}{1-\gamma} + b > 0 \right\}.$$

3 Exponential: $U(x) = -e^{-ax}$

Answer: $b = 1$ and $\gamma \rightarrow \infty$.

For $b = 1$ and $\gamma \rightarrow \infty$,

$$U(x) = \lim_{\gamma \rightarrow \infty} - \left(1 + \frac{ax}{1-\gamma} \right)^{\gamma} = -e^{\lim_{\gamma \rightarrow \infty} ax \frac{\gamma}{1-\gamma}} = -e^{-ax}.$$

Question 1 - Answer

HARA Utility Function

$$U(x) = \frac{1-\gamma}{\gamma} \left(\frac{ax}{1-\gamma} + b \right)^{\gamma}, \quad x \in \left\{ x \mid \frac{ax}{1-\gamma} + b > 0 \right\}.$$

4 Power: $U(x) = cx^{\gamma}$

Answer: $b = 0$.

For $b = 0$,

$$U(x) = \frac{(1-\gamma)^{1-\gamma} a^{\gamma}}{\gamma} x^{\gamma}.$$

Let $c = \frac{(1-\gamma)^{1-\gamma} a^{\gamma}}{\gamma}$.

Question 1 - Answer

HARA Utility Function

$$U(x) = \frac{1-\gamma}{\gamma} \left(\frac{ax}{1-\gamma} + b \right)^{\gamma}, \quad x \in \left\{ x \mid \frac{ax}{1-\gamma} + b > 0 \right\}.$$

5 Logarithmic: $U(x) = \ln(x)$

Answer: $a = 1$, $b = 0$, and $\gamma \rightarrow 0$.

For $a = 1$ and $b = 0$,

$$U(x) = \frac{(1-\gamma)^{1-\gamma}}{\gamma} x^{\gamma}.$$

Consider the equivalent form $U(x) = (x^{\gamma} - 1)/\gamma$. The for $\gamma \rightarrow 0$,

$$U(x) = \lim_{\gamma \rightarrow 0} \frac{x^{\gamma} - 1}{\gamma} = \ln x.$$

Discussions on the HARA Utility

Now we calculate the absolute risk aversion coefficient

$$A(x) = -U''(x)/U'(x).$$

- 1 Linear or risk neutral: $U(x) = x \rightarrow A(x) = 0$.
- 2 Quadratic: $U(x) = x - 0.5cx^2 \rightarrow A(x) = c/(1 - cx)$.
- 3 Exponential/CARA: $U(x) = -e^{-ax} \rightarrow A(x) = a$ (Constant Absolute Risk Aversion).
- 4 Power: $U(x) = cx^\gamma \rightarrow A(x) = (1 - \gamma)/x$.
- 5 Logarithmic: $U(x) = \ln(x) \rightarrow A(x) = 1/x$.
- 6 HARA: $U(x) = \frac{1-\gamma}{\gamma} \left(\frac{ax}{1-\gamma} + b \right)^\gamma \rightarrow A(x) = \frac{a(1-\gamma)}{ax+b(1-\gamma)}$ (Hyperbolic Absolute Risk Aversion). For $a > 0$ and $\gamma < 1$, $A(x)$ is decreasing in x .

Note that all the above utility functions have one thing in common: The Risk Tolerance

$$T(x) \equiv \frac{1}{A(x)} = Bx + C, \quad (8)$$

where B and C are constants (including ∞). That is, the risk tolerance is a **linear** function of x .

More on the HARA Utility

The HARA utility function with linear risk tolerance has many attractive properties.

- i. Optimal investment in each security is an affine function of wealth. Moreover, optimal investments in risk securities at different levels of wealth differ only by the amount but not the compositions or relative proportions.
- ii. Equilibrium allocations are Pareto optimal.
- iii. Equilibrium security prices do not depend on the distribution of agents' endowments. That is, we can construct an representative investor that assumes all the endowments.
- iv. The HARA functional form is the unique form which satisfies basic economic principles in an optimization context. Lie symmetries derive the conditions whereby the optimal solution remains invariant under scale transformations of wealth. The symmetries place restrictions on the model, with the key one being the use of HARA utility.⁷

⁷Gadi S. Perets and Eran Yashiv, 2015. The Fundamental Nature of HARA Utility, *Working Paper*, University of Lyon 1 and Tel Aviv University.

Gadi S. Perets and Eran Yashiv, 2021. Invariance in Structural Models, *Working Paper*, University of Lyon 1 and Tel Aviv University.

More Discussions on Absolute Risk Aversion

Arrow-Pratt **absolute risk aversion** coefficient is defined as

$$A(x) = -\frac{U''(x)}{U'(x)}.$$

A justification through certainty equivalent is as follows.

Let $\tilde{x}$ be a random payoff with mean $E(\tilde{x})$ and variance $Var(\tilde{x})$.
A second-order expansion near $x = E(\tilde{x})$ gives

$$U(x) \approx U(E(\tilde{x})) + U'(E(\tilde{x}))(x - E(\tilde{x})) + \frac{1}{2}U''(E(\tilde{x}))(x - E(\tilde{x}))^2. \quad (9)$$

Hence,

$$E[U(\tilde{x})] \approx U(E(\tilde{x})) + \frac{1}{2}U''(E(\tilde{x}))Var(\tilde{x}). \quad (10)$$

More Discussions on Absolute Risk Aversion

Meanwhile, let c be the certainty equivalent of $\tilde{x}$. We may regard c to be close to $E(\tilde{x})$. Then we can apply the first-order expansion to get

$$U(c) \approx U(E(\tilde{x})) + U'(E(\tilde{x}))(c - E(\tilde{x})). \quad (11)$$

Since $U(c) = E[U(\tilde{x})]$, we can solve for c using the two approximations:

$$c \approx E(\tilde{x}) + \frac{1}{2} \frac{U''(E(\tilde{x}))}{U'(E(\tilde{x}))} \text{Var}(\tilde{x}) = E(\tilde{x}) - \frac{1}{2} A(E(\tilde{x})) \text{Var}(\tilde{x}). \quad (12)$$

Note the intuitive meaning of $A(x)$ in the above formula: It measures the response of c to $\text{Var}(\tilde{x})$. The higher $A(x)$ is, the higher discount she will impose to $E(\tilde{x})$ in her personal valuation.

More Discussions on Absolute Risk Aversion

Rearranging the previous equation, the risk premium π is

$$\pi = E(\tilde{x}) - c \approx -\frac{1}{2} \frac{U''(E(\tilde{x}))}{U'(E(\tilde{x}))} \text{Var}(\tilde{x}) = \frac{1}{2} A(E(\tilde{x})) \text{Var}(\tilde{x}). \quad (13)$$

The higher $A(x)$ is, the higher risk premium she pays.

However, although concavity of the utility function, $U''(\cdot)$, determines her risk aversion, **it cannot sufficiently determine the risk premium she is willing to pay**. The risk premium also depends on her marginal utility, $U'(\cdot)$. Even if she is very risk averse (i.e., $-U''(\cdot)$ is large), she may be unwilling to pay a large risk premium if she is poor and her marginal utility is high (i.e., $U'(\cdot)$ is also large).

The Portfolio Problem



- Your total wealth levels at times 0 and 1 are W_0 and $\widetilde{W}_1$, respectively.
- There are two assets available: a riskless bond with a gross return R_f and a risky stock with a random gross return $\widetilde{R}_s$.
- You decide the amount θ_b to be invested in the riskless bond and the amount θ_s to be invested in the risky stock, such that your expected utility $E[U(\widetilde{W}_1)] = E[U(\theta_b R_f + \theta_s \widetilde{R}_s)]$ is maximized.
- In finance, portfolio choice is perhaps the most important application of expected utility theory.

The Portfolio Problem

$$\begin{aligned} \max_{\theta_b, \theta_s} \quad & E \left[U(\theta_b R_f + \theta_s \tilde{R}_s) \right], \\ \text{s.t.} \quad & \theta_b + \theta_s = W_0. \end{aligned} \tag{14}$$

Jack London would call W_0 “Gold of the Dice”.⁸

We form the Lagrangian

$$\mathcal{L} = E \left[U(\theta_b R_f + \theta_s \tilde{R}_s) \right] - \lambda (\theta_b + \theta_s - W_0), \tag{15}$$

where λ is the Lagrangian Multiplier.

⁸See his classical short story: *Love of Life*.

The Portfolio Problem: Solution

The F.O.C.s are

$$\frac{\partial \mathcal{L}}{\partial \theta_b} = E \left[U'(\theta_b^* R_f + \theta_s^* \tilde{R}_s) \right] R_f - \lambda = 0, \quad (16)$$

$$\frac{\partial \mathcal{L}}{\partial \theta_s} = E \left[U'(\theta_b^* R_f + \theta_s^* \tilde{R}_s) \tilde{R}_s \right] - \lambda = 0, \quad (17)$$

$$\frac{\partial \mathcal{L}}{\partial \lambda} = \theta_b^* + \theta_s^* - W_0 = 0. \quad (18)$$

Note that we have moved the partial derivative inside the expectation. We will provide a justification later.

The Quadratic/CARA-Normal Setup

When the utility function is either **quadratic** or **exponential** and the stock return is **normally distributed** (with mean μ and variance σ^2), we call the above setup **Quadratic-Normal** or **CARA-Normal**. This setup underlies a large class of models frequently investigated in finance.

Denote the objective function

$$f(\theta_b, \theta_s) = E \left[U(\theta_b R_f + \theta_s \tilde{R}_s) \right].$$

First, note that for quadratic or exponential utility, **dom** $f = \mathbb{R}^2$ (ignoring the satiation constraint). And the constraint set $\mathcal{C} = \{(\theta_b, \theta_s) \in \mathbb{R}^2 | \theta_b + \theta_s = W_0\}$ is affine. Hence, the feasible set $\mathcal{X} = \text{dom } f \cap \mathcal{C}$ is **convex**.

Second, recall what we have talked about in Part II. To confirm that (θ_b^*, θ_s^*) , which satisfies the F.O.C.s, is the **unique global maxima** to the portfolio problem, a sufficient condition is that f is **strictly concave**.

Sufficiency

Suppose the utility function $U(x)$ is **strictly concave** on an appropriate convex set $\mathcal{S} \subseteq \mathbb{R}$. E.g., the quadratic utility $U(x) = x - 0.5cx^2$ on $\mathbb{R}$ with $c > 0$ and the exponential utility $U(x) = -e^{-ax}$ on $\mathbb{R}$ with $a > 0$.⁹

Choose two distinct points $(\theta_b^{(1)}, \theta_s^{(1)}), (\theta_b^{(2)}, \theta_s^{(2)}) \in \mathcal{X}$. Let $(\theta_b^{(\alpha)}, \theta_s^{(\alpha)}) = \alpha (\theta_b^{(1)}, \theta_s^{(1)}) + (1 - \alpha) (\theta_b^{(2)}, \theta_s^{(2)}), 0 < \alpha < 1$.

Denote $W^{(1)} = \theta_b^{(1)} R_f + \theta_s^{(1)} R_s$, $W^{(2)} = \theta_b^{(2)} R_f + \theta_s^{(2)} R_s$, and $W^{(\alpha)} = \alpha W^{(1)} + (1 - \alpha) W^{(2)} = \theta_b^{(\alpha)} R_f + \theta_s^{(\alpha)} R_s$, where R_s is a realized value of $\tilde{R}_s$.

We have $W^{(1)} \neq W^{(2)}$ for all values of R_s , except one point

$$\bar{R}_s = -\frac{(\theta_b^{(1)} - \theta_b^{(2)}) R_f}{\theta_s^{(1)} - \theta_s^{(2)}} \quad (\text{We must have } \theta_s^{(1)} \neq \theta_s^{(2)}. \text{ Why?}).$$

⁹Try using the first- and second-order conditions to prove the strict concavity of the quadratic and exponential utility functions, respectively.

Sufficiency - Continued

Since $U(x)$ is strictly concave, for all values of R_s , except one point $\bar{R}_s = -\frac{(\theta_b^{(1)} - \theta_b^{(2)})R_f}{\theta_s^{(1)} - \theta_s^{(2)}}$, we have

$$U(W^{(\alpha)}) = U(\alpha W^{(1)} + (1 - \alpha)W^{(2)}) > \alpha U(W^{(1)}) + (1 - \alpha)U(W^{(2)}).$$

Denote $\widetilde{W}_1^{(1)} = \theta_b^{(1)}R_f + \theta_s^{(1)}\widetilde{R}_s$, $\widetilde{W}_1^{(2)} = \theta_b^{(2)}R_f + \theta_s^{(2)}\widetilde{R}_s$, and $\widetilde{W}_1^{(\alpha)} = \alpha\widetilde{W}_1^{(1)} + (1 - \alpha)\widetilde{W}_1^{(2)} = \theta_b^{(\alpha)}R_f + \theta_s^{(\alpha)}\widetilde{R}_s$. Hence,

$$\begin{aligned} f\left(\theta_b^{(\alpha)}, \theta_s^{(\alpha)}\right) &= E\left[U\left(\widetilde{W}_1^{(\alpha)}\right)\right] \\ &> \alpha E\left[U\left(\widetilde{W}_1^{(1)}\right)\right] + (1 - \alpha)E\left[U\left(\widetilde{W}_1^{(2)}\right)\right] \\ &= \alpha f\left(\theta_b^{(1)}, \theta_s^{(1)}\right) + (1 - \alpha)f\left(\theta_b^{(2)}, \theta_s^{(2)}\right). \end{aligned}$$

The **strict inequality** above requires a bit of work. See the next slide.

Sufficiency - Continued

Suppose $h(x)$ is continuous on $\mathbb{R}$ and $h(x) > 0$ for all x except that $h(\bar{x}) = 0$. Suppose further that $l(x)$ is continuous on $\mathbb{R}$ and $l(x) > 0$ for all x . Let $g(x) \equiv h(x)l(x)$. Then $g(x)$ is continuous on $\mathbb{R}$ and $g(x) > 0$ for all x except that $g(\bar{x}) = 0$.

First, choose $a < b < \bar{x}$. Then, $g(x)$ is continuous and strictly positive on $[a, b]$, and

$$\int_{-\infty}^{+\infty} g(x)dx \geq \int_a^b g(x)dx. \quad (19)$$

(Why do we have the above inequality?)

Second, applying the first mean value theorem for integration yields

$$\int_a^b g(x)dx = g(\xi)(b-a) > 0, \quad \xi \in [a, b]. \quad (20)$$

Third, write

$h(R_s) = U\left(W_1^{(\alpha)}(R_s)\right) - \left(\alpha U\left(W_1^{(1)}(R_s)\right) + (1-\alpha)U\left(W_1^{(2)}(R_s)\right)\right)$ and let $l(R_s)$ be a normal density. Then,

$$\int_{-\infty}^{+\infty} h(R_s)l(R_s)dR_s > 0. \quad (21)$$

A Pricing Relation

From (17)–(16), we get

$$E \left[U'(\widetilde{W}_1^*)(\widetilde{R}_s - R_f) \right] = 0. \quad (22)$$

Note that it holds for any risky asset and gives a **fundamental asset pricing equation**. Let $\widetilde{R}_s = \widetilde{X}_s / P_s$, where $\widetilde{X}_s$ is the total random payoff at time 1 and P_s is the price of the risky asset at time 0. Then

$$P_s = \frac{1}{R_f} \frac{E[U'(\widetilde{W}_1^*)\widetilde{X}_s]}{E[U'(\widetilde{W}_1^*)]} \equiv f(P_s). \quad (23)$$

Thus, P_s is a fixed point of f .

(We have not completely solved the pricing problem since we still do not know what R_f is. This requires the market clearing conditions of a general equilibrium to settle.)

Optimal Portfolios: Comparative Statics

The budget constraint in (18) imposes the following restrictions.

$$\frac{\partial \theta_s^*}{\partial W_0} + \frac{\partial \theta_b^*}{\partial W_0} = 1, \quad (24)$$

$$\frac{\partial \theta_s^*}{\partial \mu} + \frac{\partial \theta_b^*}{\partial \mu} = \frac{\partial \theta_s^*}{\partial \sigma} + \frac{\partial \theta_b^*}{\partial \sigma} = \frac{\partial \theta_s^*}{\partial R_f} + \frac{\partial \theta_b^*}{\partial R_f} = 0, \quad (25)$$

$$\frac{\partial \theta_s^*}{\partial \varsigma} + \frac{\partial \theta_b^*}{\partial \varsigma} = 0, \quad (26)$$

where ς is any parameter of the utility function.

Therefore, any parameter other than W_0 affects the optimal investments in the risky and riskfree assets in **exactly opposite** directions.

Optimal Portfolios under Quadratic Utility

For the quadratic utility $U(x) = x - 0.5cx^2$, $c > 0$,

$$\begin{aligned} E(U(\tilde{x})) &= E(\tilde{x} - 0.5c\tilde{x}^2) \\ &= E(\tilde{x}) - 0.5c(E(\tilde{x}))^2 - 0.5c\text{Var}(\tilde{x}). \end{aligned}$$

This can be seen to be **equivalent to a mean-variance approach in the following sense.**

- 1 For convenience, suppose the initial wealth is 1. Then $\tilde{x}$ corresponds to the return $\tilde{R}$. Fix the expected value of the solution, i.e., $E(\tilde{x}^*) = \mu$. Then, $\tilde{x}$ must have minimum variance for all feasible $\tilde{x}$ satisfying $E(\tilde{x}) = \mu$.
- 2 Since $\tilde{x} = \tilde{R}$, it follows that the solution must correspond to a mean-variance efficient point.

Optimal Portfolios under Quadratic Utility

For the quadratic utility $U(x) = x - 0.5cx^2$, $c > 0$, the F.O.C.s are

$$\frac{\partial \mathcal{L}}{\partial \theta_b} = R_f - cR_f^2\theta_b^* - c\mu R_f\theta_s^* - \lambda = 0, \quad (27)$$

$$\frac{\partial \mathcal{L}}{\partial \theta_s} = \mu - c\mu R_f\theta_b^* - c(\sigma^2 + \mu^2)\theta_s^* - \lambda = 0, \quad (28)$$

$$\frac{\partial \mathcal{L}}{\partial \lambda} = \theta_b^* + \theta_s^* - W_0 = 0. \quad (29)$$

Optimal Portfolios under Quadratic Utility

$$\theta_s^* = \frac{\mu - R_f}{c(\sigma^2 + (\mu - R_f)^2)} - \frac{(\mu - R_f)R_f}{\sigma^2 + (\mu - R_f)^2} W_0, \quad (30)$$

$$\theta_b^* = -\frac{\mu - R_f}{c(\sigma^2 + (\mu - R_f)^2)} + \frac{\sigma^2 + \mu^2 - \mu R_f}{\sigma^2 + (\mu - R_f)^2} W_0. \quad (31)$$

Recall that for quadratic utility, $A(x) = c/(1 - cx)$. Hence, $c = A/(1 + Ax) = 1/(A^{-1} + x)$.

Given x , a higher A corresponds to a higher c . Therefore, we can study the effect of risk aversion on the optimal portfolio through c .

Clearly, for both θ_s^* and θ_b^* , only the first term is affected by c , and only the 2nd term is affected by W_0 .

Optimal Portfolios under Quadratic Utility

Consider the realistic case of $\mu > R_f$. We have

$$1 \quad \frac{\partial \theta_s^*}{\partial c} = -\frac{\mu - R_f}{c^2(\sigma^2 + (\mu - R_f)^2)} < 0, \quad \frac{\partial \theta_b^*}{\partial c} = -\frac{\partial \theta_s^*}{\partial c} > 0.$$

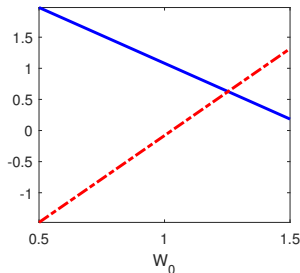
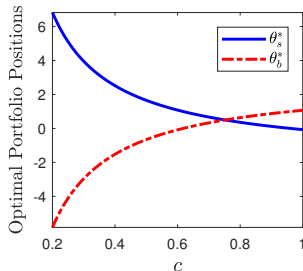
- Intuition: An investor armed with a quadratic utility reduces her investment in the risky asset and increases her investment in the safe asset, when she becomes more risk-averse.

$$2 \quad \frac{\partial \theta_s^*}{\partial W_0} = -\frac{(\mu - R_f)R_f}{\sigma^2 + (\mu - R_f)^2} < 0, \quad \frac{\partial \theta_b^*}{\partial W_0} = 1 - \frac{\partial \theta_s^*}{\partial W_0} > 1.$$

- This result is a bit **counterintuitive**. We may expect θ_s^* not to decrease for a higher W_0 .
- Since $A(x) = c/(1 - cx)$, an investor armed with a quadratic utility becomes more risk averse when she gets wealthier (i.e., a higher x). This effect **dominates** and diminishes her risky position.

A Numerical Example

Choose the following baseline parameters: $c = 0.60$, $\mu = 1.12$, $\sigma = 0.2$, $R_f = 1.04$, and $W_0 = 1$.



Optimal Portfolios under Quadratic Utility

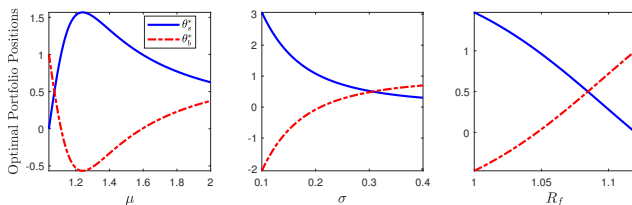
How about the other three parameters, μ , σ , and R_f ? We rewrite θ_s^* as

$$\theta_s^* = \frac{1}{\frac{\sigma^2}{\mu - R_f} + (\mu - R_f)} \times (1/c - R_f W_0). \quad (32)$$

- Given σ , the left term is maximized when $\mu - R_f = \sigma$, and it decreases when $\mu - R_f$ moves away from σ . We expect a nonmonotonic relation between θ_s^* and μ , with a turning point at $R_f + \sigma$. If the sign of the right term is positive, θ_s^* is first increasing and then decreasing w.r.t. μ .
- When the sign of the right term is positive, we can easily see that θ_s^* is decreasing w.r.t. σ .
- For $\sigma > \mu - R_f > 0$, the left term is decreasing in R_f . Ignoring the effect of R_f in the right term, we find that θ_s^* is decreasing in R_f when the right term is positive.

A Numerical Example

Choose the following baseline parameters: $c = 0.60$, $\mu = 1.12$, $\sigma = 0.2$, $R_f = 1.04$, and $W_0 = 1$.



Optimal Portfolios under Quadratic Utility

Intuition for μ :

- Given W_0 , we may expect the investor to increase her investment in the risky asset when the risky asset becomes more attractive — providing a higher expected return.
- However, we find a nonmonotonic relation between θ_s^* and μ . The reason is that μ also brings about a wealth effect which in turn impacts the investor's risk aversion and exerts an opposite influence on the optimal investments. The ultimate outcome is a tradeoff.
- Specifically, a higher expected return of the risky asset increases her expected wealth level, which simultaneously lifts her risk aversion. This provides a motivation for her to reduce her risky asset position. The total effect depends on the detailed parameter values.
- Recall that the turning point is $\mu = R_f + \sigma$, which is not related to c . Note also that the satiation point of the quadratic utility, $1/c$, is determined solely by c . So what is the role played by satiation?

The Effect of Satiation

Since $A(x) = \frac{1}{1/c-x} \rightarrow \infty$ when $x \rightarrow (1/c)-$, we may speculate that the investor will not allow her expected future wealth to exceed the satiation point $1/c$, after which her utility actually starts to decline.

Using the optimal solutions, we have

$$\begin{aligned} E(\widetilde{W}_1^*) &= E(\theta_b^* R_f + \theta_s^* \widetilde{R}_s) = \theta_b^* R_f + \theta_s^* \mu \\ &= \frac{1}{c} - \frac{\sigma^2}{\sigma^2 + (\mu - R_f)^2} \left(\frac{1}{c} - R_f W_0 \right). \end{aligned}$$

Under the mild condition that even if the investor puts all her money in the riskfree asset, her future wealth will not reach the satiation point, that is, $R_f W_0 < 1/c$, we obtain

$$E(\widetilde{W}_1^*) < \frac{1}{c},$$

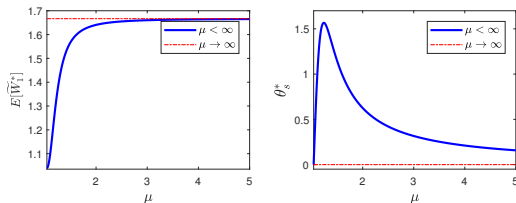
and when $\mu \rightarrow \infty$,

$$E(\widetilde{W}_1^*) \rightarrow \frac{1}{c} \text{ and } \theta_s^* \rightarrow 0.$$

Since $\theta_s^* \rightarrow 0$ when $\mu \rightarrow R_f$, we know that θ_s^* must first rise but eventually fall with μ .

A Numerical Example

Choose the following baseline parameters: $c = 0.60$, $\mu = 1.12$, $\sigma = 0.2$, $R_f = 1.04$, and $W_0 = 1$. The satiation point is $E(\widetilde{W}_1^*) \rightarrow \frac{1}{c} = 1.67$.



Optimal Portfolios under Exponential Utility

For the exponential utility $U(x) = -e^{-ax}$, $a > 0$, the F.O.C.s are

$$\frac{\partial \mathcal{L}}{\partial \theta_b} = aR_f e^{-aR_f \theta_b^*} E[e^{-a\theta_s^* \tilde{R}_s}] - \lambda = 0,$$

$$\frac{\partial \mathcal{L}}{\partial \theta_s} = a e^{-aR_f \theta_b^*} E[\tilde{R}_s e^{-a\theta_s^* \tilde{R}_s}] - \lambda = 0,$$

$$\frac{\partial \mathcal{L}}{\partial \lambda} = \theta_b^* + \theta_s^* - W_0 = 0.$$

Let $f(x, \mu, \sigma^2)$ be the density of $x \sim N(\mu, \sigma^2)$. We have

$$\begin{aligned} E[\tilde{R}_s e^{-a\theta_s^* \tilde{R}_s}] &= \int_{-\infty}^{+\infty} R_s e^{-a\theta_s^* R_s} f(R_s, \mu, \sigma^2) dR_s \\ &= e^{-a\mu\theta_s^* + 0.5a^2\sigma^2\theta_s^{*2}} \int_{-\infty}^{+\infty} R_s f(R_s, \mu - a\sigma^2\theta_s^*, \sigma^2) dR_s \\ &= (\mu - a\sigma^2\theta_s^*) e^{-a\mu\theta_s^* + 0.5a^2\sigma^2\theta_s^{*2}}. \end{aligned}$$

Optimal Portfolios under Exponential Utility

$$aR_f e^{-aR_f\theta_b^*} e^{-a\mu\theta_s^* + 0.5a^2\sigma^2\theta_s^{*2}} = \lambda, \quad (33)$$

$$\implies a(\mu - a\sigma^2\theta_s^*) e^{-aR_f\theta_b^*} e^{-a\mu\theta_s^* + 0.5a^2\sigma^2\theta_s^{*2}} = \lambda, \quad (34)$$

$$\theta_b^* + \theta_s^* = W_0. \quad (35)$$

(34)÷(33), we get

$$\theta_s^* = \frac{\mu - R_f}{a\sigma^2}, \quad \theta_b^* = W_0 - \frac{\mu - R_f}{a\sigma^2}. \quad (36)$$

Very intuitively, an investor increases her position in the risky asset when

- its risk premium is higher (i.e., a higher μ and/or a lower R_f);
- its risk is lower (i.e., a smaller σ);
- she has lower absolute risk aversion (i.e., a lower $A(x) = a$).

Question 2

Note that $E[U(\tilde{x})] = -E[e^{-a\tilde{x}}] = -e^{-aE[\tilde{x}] + 0.5a^2 \text{Var}[\tilde{x}]}$. Hence,

$$\max E[U(\tilde{x})] \iff \max E[\tilde{x}] - 0.5a \text{Var}[\tilde{x}].$$

Using this result to re-derive the optimal portfolio weights.

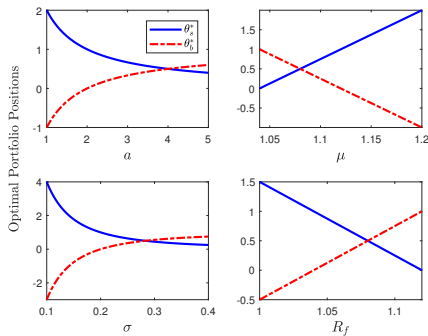
The utility function $E[\tilde{x}] - 0.5a \text{Var}[\tilde{x}]$ is also called a **derived utility** and the name is self-explanatory. Note that it depends on both the form of the utility function and the return distribution. Moreover, it looks similar to the expected quadratic utility ($E[\tilde{x}] - 0.5aE[\tilde{x}]^2 - 0.5a \text{Var}[\tilde{x}]$). However, the missing of the $E[\tilde{x}]^2$ term makes the portfolio results very different.¹⁰

¹⁰Actually, $E[\tilde{x}] - 0.5a \text{Var}[\tilde{x}]$ is the mean-variance **criterion** popularly used in the practice of portfolio management. However, it **cannot** be written as an expected utility and hence is hard to use in a dynamic portfolio choice environment. See the following two references for more discussions.

- Basak, S. and G. Chabakauri, 2010. Dynamic Mean-Variance Asset Allocation, *Review of Financial Studies* 23, 2970-3016.
- Luo, D., B. Shi, and H. Yao, 2024. Intertemporal Capital Asset Pricing: A Continuous-Time Mean-Variance Approach, *Working Paper*.

A Numerical Example

Choose the following baseline parameters: $a = 2$, $\mu = 1.12$, $\sigma = 0.2$, $R_f = 1.04$, and $W_0 = 1$.

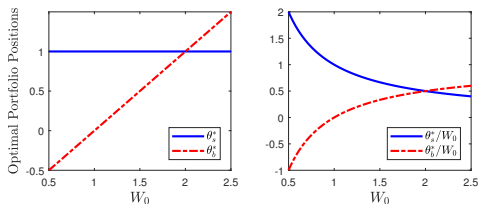


The Effect of Initial Wealth

An interesting observation:

$$\frac{\partial \theta_s^*}{\partial W_0} = 0, \quad \frac{\partial \theta_b^*}{\partial W_0} = 1.$$

An investor armed with the exponential utility function invests **a fixed amount** in the risky asset no matter her initial wealth level.



Optimal Portfolios under Power Utility

For the Power utility $U(x) = \frac{x^{1-\gamma}}{1-\gamma}$, $x > 0$, $\gamma > 0$ and $\gamma \neq 1$, we make the following changes and introduce the **CRRA-Lognormal setup**.¹¹

We write $\widetilde{W}_1 = \theta_b R_f + \theta_s \widetilde{R}_s = W_0(\phi_b R_f + \phi_s \widetilde{R}_s)$, where ϕ_b and ϕ_s are the **fractions of wealth** invested in the riskfree and risky assets, respectively.

We further approximate $\widetilde{W}_1$ by a **lognormal distribution** thus $\widetilde{W}_1 \approx W_0 e^{\phi_b r_f + \phi_s \widetilde{r}_s - 0.5 \phi_s^2 \sigma_s^2}$, where r_f is the constant riskfree interest rate and $\widetilde{r}_s \sim N(\mu_s, \sigma_s^2)$ is the random return of the risky asset. This ensures that $\widetilde{W}_1 > 0$.

The term $-0.5 \phi_s^2 \sigma_s^2$ is called the Itô correction term, also called the convexity adjustment term. Note that $E[\widetilde{W}_1] \approx W_0 e^{\phi_b r_f + \phi_s \mu_s}$. The expected (log) growth rate of W is $\phi_b r_f + \phi_s \mu_s$, as desired.

¹¹The meaning of “CRRA” will be explained soon.

Quadratic/CARA-Normal v.s. CRRA-Lognormal

- In the Quadratic/CARA-Normal setup, $\widetilde{W}_1 = \theta_b R_f + \theta_s \widetilde{R}_s$ is **normally** distributed.
- In the CRRA-LogNormal setup, $\widetilde{W}_1 \approx W_0 e^{\phi_b r_f + \phi_s \widetilde{r}_s - 0.5 \phi_s^2 \sigma_s^2}$ is **lognormally** distributed.

Note that in the latter, we only get an **approximate** result because, as we discussed in Part I, the log return of a portfolio equals the value-weight average of the log returns of the underlying assets only approximately.

Lognormal Approximation

However, this approximation is caused by the buy-and-hold strategy we have adopted so far, and can be made exact through **continuous portfolio rebalancing**. To be precise, we treat time as a continuum, and adjust our portfolios every instant such that the weights remain at ϕ_b and ϕ_s at any time.

Heuristically, let's divide the single time period into T small time intervals, each with length $\Delta t = 1/T$. Because of the rebalancing at the beginning of each period,

$$\tilde{W}_{i\Delta t} \approx \tilde{W}_{(i-1)\Delta t} e^{\phi_b r_f \Delta t + \phi_s \tilde{r}_{s,i} \Delta t - 0.5 \phi_s^2 \sigma_s^2 \Delta t}, \quad i = 1, 2, \dots, T,$$

where $\tilde{r}_{s,i} \Delta t \sim i.i.d. N(\mu_s \Delta t, \sigma_s^2 \Delta t)$ and $\tilde{W}_0 \equiv W_0$. Applying second-order expansion to the exponential, and noting that $\tilde{r}_{s,i}^2 \Delta t^2$ is actually of order Δt , that is, $\tilde{r}_{s,i}^2 \Delta t^2 = O(\Delta t)$ or $\tilde{r}_{s,i}^2 \Delta t^2 \sim \sigma_s^2 \Delta t$, we obtain

$$\begin{aligned} \tilde{W}_{i\Delta t} &\approx \tilde{W}_{(i-1)\Delta t} (1 + \phi_b r_f \Delta t + \phi_s \tilde{r}_{s,i} \Delta t - 0.5 \phi_s^2 \sigma_s^2 \Delta t + 0.5 \phi_s^2 \sigma_s^2 \Delta t + o(\Delta t)), \\ &\approx \tilde{W}_{(i-1)\Delta t} (1 + \phi_b r_f \Delta t + \phi_s \tilde{r}_{s,i} \Delta t + o(\Delta t)), \quad i = 1, 2, \dots, T. \end{aligned}$$

Then

$$\begin{aligned} \tilde{W}_1 = \tilde{W}_{T\Delta t} &\approx W_0 \prod_{i=1}^T (1 + \phi_b r_f \Delta t + \phi_s \tilde{r}_{s,i} \Delta t + o(\Delta t)) \\ &\approx W_0 \left(1 + \phi_b r_f \sum_{i=1}^T \Delta t + \phi_s \sum_{i=1}^T \tilde{r}_{s,i} \Delta t + o(\Delta t) \right) \\ &= W_0 (1 + \phi_b r_f + \phi_s \tilde{r}_s + o(\Delta t)) \\ &\approx W_0 e^{\phi_b r_f + \phi_s \tilde{r}_s - 0.5 \phi_s^2 \sigma_s^2}. \end{aligned}$$

Let $\Delta t \rightarrow 0+$. We have $\tilde{W}_1 = W_0 e^{\phi_b r_f + \phi_s \tilde{r}_s - 0.5 \phi_s^2 \sigma_s^2}$.

The Portfolio Problem

$$\begin{aligned} \max_{\phi_b, \phi_s} \quad & E \left[U(W_0 e^{\phi_b r_f + \phi_s \tilde{r}_s - 0.5 \phi_s^2 \sigma_s^2}) \right], \\ \text{s.t.} \quad & \phi_b + \phi_s = 1. \end{aligned} \quad (37)$$

We form the Lagrangian

$$\mathcal{L} = E \left[U(W_0 e^{\phi_b r_f + \phi_s \tilde{r}_s - 0.5 \phi_s^2 \sigma_s^2}) \right] - \lambda (\phi_b + \phi_s - 1), \quad (38)$$

where λ is the Lagrangian Multiplier.

Optimal Portfolios under Power Utility

Since

$$E \left[U(W_0 e^{\phi_b r_f + \phi_s \tilde{r}_s - 0.5 \phi_s^2 \sigma_s^2}) \right] = \frac{W_0^{1-\gamma}}{1-\gamma} e^{(1-\gamma)(\phi_b r_f + \phi_s \mu_s - 0.5 \gamma \phi_s^2 \sigma_s^2)},$$

the F.O.C.s are

$$\frac{\partial \mathcal{L}}{\partial \phi_b} = W_0^{1-\gamma} r_f e^{(1-\gamma)(r_f \phi_b^* + \mu_s \phi_s^* - 0.5 \gamma \sigma_s^2 \phi_s^{*2})} - \lambda = 0,$$

$$\frac{\partial \mathcal{L}}{\partial \phi_s} = W_0^{1-\gamma} (\mu_s - \gamma \sigma_s^2 \phi_s^*) e^{(1-\gamma)(r_f \phi_b^* + \mu_s \phi_s^* - 0.5 \gamma \sigma_s^2 \phi_s^{*2})} - \lambda = 0,$$

$$\frac{\partial \mathcal{L}}{\partial \lambda} = \phi_b^* + \phi_s^* - 1 = 0.$$

Optimal Portfolios under Power Utility

$$W_0^{1-\gamma} r_f e^{(1-\gamma)(r_f \phi_b^* + \mu_s \phi_s^* - 0.5 \gamma \sigma_s^2 \phi_s^{*2})} = \lambda, \quad (39)$$

$$\implies W_0^{1-\gamma} (\mu_s - \gamma \sigma_s^2 \phi_s^*) e^{(1-\gamma)(r_f \phi_b^* + \mu_s \phi_s^* - 0.5 \gamma \sigma_s^2 \phi_s^{*2})} = \lambda, \quad (40)$$

$$\phi_b^* + \phi_s^* = 1. \quad (41)$$

(40) ÷ (39), we get

$$\phi_s^* = \frac{\mu_s - r_f}{\gamma \sigma_s^2}, \quad \phi_b^* = 1 - \frac{\mu_s - r_f}{\gamma \sigma_s^2}. \quad (42)$$

Very intuitively, an investor increases her position in the risky asset when

- its risk premium is higher (i.e., a higher μ_s and/or a lower r_f);
- its risk is lower (i.e., a smaller σ_s);
- she is less risk averse (i.e., a lower $A(x) = \gamma/x$, or a lower γ given x).

The CRRA-LogNormal Setup

For power utility with $\gamma > 0$ and $\gamma \neq 1$,

$$E \left[U(\widetilde{W}_1) \right] = \frac{W_0^{1-\gamma}}{1-\gamma} e^{(1-\gamma)(\phi_b r_f + \phi_s \mu_s - 0.5\gamma \phi_s^2 \sigma_s^2)}.$$

Hence,

$$\max_{\phi_b, \phi_s} E \left[U(\widetilde{W}_1) \right] \Leftrightarrow \max_{\phi_b, \phi_s} \phi_b r_f + \phi_s \mu_s - 0.5\gamma \phi_s^2 \sigma_s^2.$$

Let $f(\phi_b, \phi_s) = r_f \phi_b + \mu_s \phi_s - 0.5\gamma \phi_s^2 \sigma_s^2$.

First, note that **dom** $f = \mathbb{R}^2$. And the constraint set $\mathcal{C}$ is affine. Hence, the feasible set $\mathcal{X} = \mathbf{dom} \, f \cap \mathcal{C}$ is **convex**.

Second, recall what we have talked about in Part II. To confirm that (ϕ_b^*, ϕ_s^*) , which satisfies the F.O.C.s, is **a unique global maxima** to the portfolio problem, a sufficient condition is that f is **strictly concave**.

Sufficiency

$$f(\phi_b, \phi_s) = r_f \phi_b + \mu_s \phi_s - 0.5 \gamma \sigma_s^2 \phi_s^2.$$

Note that $\nabla^2 f = \begin{bmatrix} 0 & 0 \\ 0 & -\gamma \sigma_s^2 \end{bmatrix}$ is negative semi-definite. Hence we can conclude that f is concave but **not necessarily** strictly concave. We turn to first-order conditions for strict concavity. We cannot directly discuss concavity on $\mathcal{X}$, because it is not open. Now we try concavity on $\mathbb{R}^2$.

For any two distinct points $(\phi_b^{(1)}, \phi_s^{(1)}), (\phi_b^{(2)}, \phi_s^{(2)}) \in \mathbb{R}^2$, we have

$$\begin{aligned} f(\phi_b^{(1)}, \phi_s^{(1)}) - f(\phi_b^{(2)}, \phi_s^{(2)}) - \nabla f(\phi_b^{(2)}, \phi_s^{(2)}) \left((\phi_b^{(1)}, \phi_s^{(1)}) - (\phi_b^{(2)}, \phi_s^{(2)}) \right)^\top \\ &= f(\phi_b^{(1)}, \phi_s^{(1)}) - f(\phi_b^{(2)}, \phi_s^{(2)}) - (r_f, \mu_s - \gamma \sigma_s^2 \phi_s^{(2)}) (\phi_b^{(1)} - \phi_b^{(2)}, \phi_s^{(1)} - \phi_s^{(2)})^\top \\ &= -0.5 \gamma \sigma_s^2 (\phi_s^{(1)})^2 + 0.5 \gamma \sigma_s^2 (\phi_s^{(2)})^2 + \gamma \sigma_s^2 \phi_s^{(2)} (\phi_s^{(1)} - \phi_s^{(2)}) \\ &= -0.5 \gamma \sigma_s^2 \left((\phi_s^{(1)})^2 - (\phi_s^{(2)})^2 - 2 \phi_s^{(1)} \phi_s^{(2)} + 2 (\phi_s^{(2)})^2 \right) \\ &= -0.5 \gamma \sigma_s^2 (\phi_s^{(1)} - \phi_s^{(2)})^2 \leq 0. \end{aligned}$$

Therefore, f is again concave but **not necessarily** strictly concave.

Sufficiency - Continued

We are left with the last choice of proof by definition.

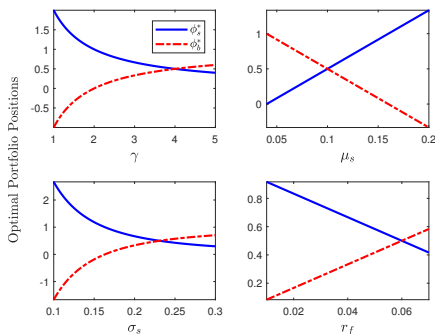
For any two distinct points $(\phi_b^{(1)}, \phi_s^{(1)}), (\phi_b^{(2)}, \phi_s^{(2)}) \in \mathcal{X}$, we must have $\phi_s^{(1)} \neq \phi_s^{(2)}$. Let $(\phi_b^\alpha, \phi_s^\alpha) = \alpha(\phi_b^{(1)}, \phi_s^{(1)}) + (1 - \alpha)(\phi_b^{(2)}, \phi_s^{(2)})$, $0 < \alpha < 1$. Then,

$$\begin{aligned} f(\phi_b^\alpha, \phi_s^\alpha) &- \left(\alpha f(\phi_b^{(1)}, \phi_s^{(1)}) + (1 - \alpha) f(\phi_b^{(2)}, \phi_s^{(2)}) \right) \\ &= 0.5\gamma\sigma_s^2\alpha(1 - \alpha)(\phi_s^{(1)} - \phi_s^{(2)})^2 > 0. \end{aligned}$$

Therefore, f is **strictly concave** on $\mathcal{X}$.

A Numerical Example

Choose the following baseline parameters: $\gamma = 3$, $\mu_s = 0.12$, $\sigma_s = 0.2$, $r_f = 0.04$, and $W_0 = 1$.

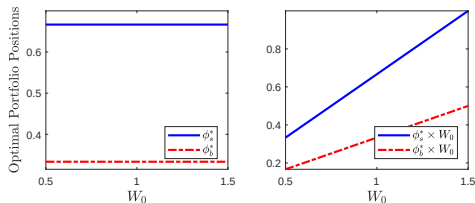


The Effect of Initial Wealth

An interesting observation:

$$\frac{\partial \phi_s^*}{\partial W_0} = \frac{\partial \phi_b^*}{\partial W_0} = 0.$$

An investor armed with the power utility function invests **a fixed fraction** in the risky asset no matter her initial wealth level.



Question 3

Optimal Portfolios under Log Utility

When $\gamma \rightarrow 1$, we get the log utility $U(x) = \ln(x)$. Solve the previous portfolio problem under log utility.

Question 3: Answer

Answer: For log utility,

$$E \left[U(\widetilde{W}_1) \right] = \ln W_0 + \phi_b r_f + \phi_s \mu_s - 0.5 \phi_s^2 \sigma_s^2.$$

Define

$$\mathcal{L} = \ln W_0 + \phi_b r_f + \phi_s \mu_s - 0.5 \phi_s^2 \sigma_s^2 - \lambda (\phi_b + \phi_s - 1), \quad (43)$$

where λ is the Lagrangian Multiplier. The F.O.C.s are

$$\begin{aligned} \frac{\partial \mathcal{L}}{\partial \phi_b} &= r_f - \lambda = 0, \\ \frac{\partial \mathcal{L}}{\partial \phi_s} &= \mu_s - \sigma_s^2 \phi_s^* - \lambda = 0, \\ \frac{\partial \mathcal{L}}{\partial \lambda} &= \phi_b^* + \phi_s^* - 1 = 0. \end{aligned}$$

Finally, we get

$$\phi_s^* = \frac{\mu_s - r_f}{\sigma_s^2}, \quad \phi_b^* = 1 - \frac{\mu_s - r_f}{\sigma_s^2}. \quad (44)$$

Relative Risk Aversion

The above results motivate us to define another measure of risk aversion: the **Relative Risk Aversion Coefficient**

$$R(x) = xA(x) = -\frac{xU''(x)}{U'(x)}.$$

According to our previous discussions, we can see that absolute risk aversion indicates how the investor's dollar amount in the risky asset changes with changes in initial wealth while relative risk aversion indicates how the investor's portfolio proportion (or portfolio weight) in the risky asset changes with changes in initial wealth.

The power utility function has a **Constant Relative Risk Aversion Coefficient**. Hence it is often called a **CRRA utility function**.

Question 4

Optimal Portfolios under HARA Utility

Consider the simplified HARA utility $U(x) = \frac{(x-c)^{1-\gamma}}{1-\gamma}$,

$0 < \gamma < 1$, $c > 0$ and $x - c \geq 0$. The terminal wealth $\widetilde{W}_1$ follows a lognormal distribution, i.e., $\widetilde{W}_1 = W_0 e^{\phi_b r_f + \phi_s \tilde{r}_s - 0.5 \phi_s^2 \sigma_s^2}$, where r_f is the constant riskfree interest rate and $\tilde{r}_s \sim N(\mu_s, \sigma_s^2)$ is the random return of the risky asset. What are the F.O.C.s of this portfolio problem (ignoring the constraint $x \geq c$)? Can we solve them? (How to deal with the case $\gamma > 1$ and $x - c > 0$? Can we ignore the constraint $x - c > 0$?)

Question 4: Answer

Answer: The F.O.C.s are

$$\frac{\partial \mathcal{L}}{\partial \phi_b} = E \left[\left(W_0 e^{\phi_b^* r_f + \phi_s^* \tilde{r}_s - 0.5 \phi_s^{*2} \sigma_s^2} - c \right)^{-\gamma} W_0 e^{\phi_b^* r_f + \phi_s^* \tilde{r}_s - 0.5 \phi_s^{*2} \sigma_s^2} \right] r_f - \lambda = 0,$$

$$\frac{\partial \mathcal{L}}{\partial \phi_s} = E \left[\left(W_0 e^{\phi_b^* r_f + \phi_s^* \tilde{r}_s - 0.5 \phi_s^{*2} \sigma_s^2} - c \right)^{-\gamma} W_0 e^{\phi_b^* r_f + \phi_s^* \tilde{r}_s - 0.5 \phi_s^{*2} \sigma_s^2} (\tilde{r}_s - \sigma_s^2 \phi_s^*) \right] - \lambda = 0,$$

$$\frac{\partial \mathcal{L}}{\partial \lambda} = \phi_b^* + \phi_s^* - 1 = 0.$$

These equations are generally **insolvable**. Four ways lie ahead:

- 1 We may consider the special case of $U(x) = \ln(x - c)$ by letting $\gamma \rightarrow 1$.
- 2 We may carry out the expectation with numerical integration.
- 3 We may approximate the integral to achieve analytical solutions.
- 4 We may formulate a corresponding continuous-time model which admits analytical solutions.

Investment Horizon

We can also say something about investment horizon.

— Power: $\phi_S^* = \frac{\mu_S - r_f}{\gamma \sigma_S^2}$.

For investment horizon of T instead of 1, we simply replace $\mu_S - r_f$ with $(\mu_S - r_f)T$ and σ_S^2 with $\sigma_S^2 T$.

— Power: $\phi_S^*(T) = \frac{(\mu_S - r_f)T}{\gamma \sigma_S^2 T} = \frac{\mu_S - r_f}{\gamma \sigma_S^2}$.

The optimal fraction of wealth in the risky asset does **not** increase with investment horizon. This seems inconsistent with conventional wisdom that investors take more risk for a longer investment horizon (e.g., “put 100-minus-your-age percent of your wealth in stocks”). A possible explanation is based on return predictability in a multi-period model.

Taking Derivatives under the Integral Sign

Theorem 3.4 of Lang (1997)

Let f be a function of two variables x, y defined for $x \geq a$ and for y in some closed interval $\mathcal{I} = [c, d]$, $c < d$. Assume that $\frac{\partial f}{\partial y}$ exists, and that both f and $\frac{\partial f}{\partial y}$ are continuous. Assume that

$$\int_a^\infty \frac{\partial f(x, y)}{\partial y} dx \quad (45)$$

converges uniformly for $y \in \mathcal{I}$, and that

$$g(y) = \int_a^\infty f(x, y) dx \quad (46)$$

converges for all y . Then g is differentiable, and

$$g'(y) = \int_a^\infty \frac{\partial f(x, y)}{\partial y} dx. \quad (47)$$

The Quadratic-Normal Case

Let $g(\theta_b) \equiv E \left[U(\theta_b R_f + \theta_s \tilde{R}_s) \right]$, where $U(x) = x - 0.5cx^2$, $c > 0$, and $\tilde{R}_s \sim N(\mu_s, \sigma_s^2)$ with density $f(R_s, \mu_s, \sigma_s^2)$. Then

$$g(\theta_b) = \int_{-\infty}^{\infty} U(\theta_b R_f + \theta_s R_s) f(R_s, \mu_s, \sigma_s^2) dR_s. \quad (48)$$

For any θ_b and θ_s , we can find a sufficiently large $l > 0$, such that $\theta_b, \theta_s \in [-l, l]$.

Obviously, $g(\theta_b)$ converges for all θ_b .

We can write

$$\begin{aligned} g(\theta_b) &= \int_a^{\infty} U(\theta_b R_f + \theta_s R_s) f(R_s, \mu_s, \sigma_s^2) dR_s \\ &\quad + \int_{-a}^{\infty} U(\theta_b R_f - \theta_s R_s) f(-R_s, \mu_s, \sigma_s^2) dR_s, \end{aligned}$$

where a is an arbitrary real number. The two integrals above converge for all θ_b .

Denote $g_1(\theta_b) = R_f \int_a^{\infty} U'(\theta_b R_f + \theta_s R_s) f(R_s, \mu_s, \sigma_s^2) dR_s$. We now show that $g_1(\theta_b)$ converges uniformly for $\theta_b \in [-l, l]$.

The Quadratic-Normal Case - Cont'd

We have

$$g_1(\theta_b) = R_f \int_a^\infty (1 - c\theta_b R_f - c\theta_s R_s) f(R_s, \mu_s, \sigma_s^2) dR_s.$$

We need to show that for any $\varepsilon > 0$, there exists b' , such that for all $b > b'$,

$$R_f \left| \int_b^\infty (1 - c\theta_b R_f - c\theta_s R_s) f(R_s, \mu_s, \sigma_s^2) dR_s \right| < \varepsilon, \text{ for } \theta_b \in [-l, l].$$

It is innocuous to assume that $b' > 1 + \mu_s$. Then,

$$\begin{aligned} & R_f \left| \int_b^\infty (1 - c\theta_b R_f - c\theta_s R_s) f(R_s, \mu_s, \sigma_s^2) dR_s \right| \\ & \leq R_f \int_b^\infty |1 - c\theta_b R_f - c\theta_s R_s| f(R_s, \mu_s, \sigma_s^2) dR_s \\ & = R_f \int_{b-\mu_s}^\infty |1 - c\theta_b R_f - c\theta_s \mu_s - c\theta_s R_s| f(R_s, 0, \sigma_s^2) dR_s \\ & \leq R_f \int_{b-\mu_s}^\infty (1 + c(R_f + |\mu_s|) + cR_s) f(R_s, 0, \sigma_s^2) dR_s \\ & = \frac{R_f}{\sqrt{2\pi}\sigma_s} \int_{b-\mu_s}^\infty \frac{1 + c(R_f + |\mu_s|) + cR_s}{\frac{R_s^2}{e^{2\sigma_s^2}}} dR_s \end{aligned}$$

The Quadratic-Normal Case - Cont'd

$$\begin{aligned}
 &\leq \frac{2R_f\sigma_s^2(1 + cl(R_f + |\mu_s|))}{\sqrt{2\pi}\sigma_s} \int_{b-\mu_s}^{\infty} \frac{1}{R_s^2} dR_s + \frac{8R_f\sigma_s^4 cl}{\sqrt{2\pi}\sigma_s} \int_{b-\mu_s}^{\infty} \frac{1}{R_s^3} dR_s \\
 &= \frac{2R_f\sigma_s(1 + cl(R_f + |\mu_s|))}{\sqrt{2\pi}} \frac{1}{b - \mu_s} + \frac{4R_f\sigma_s^3 cl}{\sqrt{2\pi}} \frac{1}{(b - \mu_s)^2} \\
 &\leq \frac{2R_f\sigma_s(1 + cl(R_f + |\mu_s|)) + 4R_f\sigma_s^3 cl}{\sqrt{2\pi}} \frac{1}{b - \mu_s}
 \end{aligned}$$

Thus, we can set $b' = \max\{1 + \mu_s, \frac{2R_f\sigma_s(1+cl(R_f+|\mu_s|))+4R_f\sigma_s^3 cl}{\sqrt{2\pi}} \frac{1}{\varepsilon} + \mu_s\}$.

We can similarly define $g_2(\theta_b) = R_f \int_{-a}^{\infty} U'(\theta_b R_f - \theta_s R_s) f(-R_s, \mu_s, \sigma_s^2) dR_s$, and show that $g_2(\theta_b)$ converges uniformly for $\theta_b \in [-l, l]$.

Finally, by Theorem 3.4 of Lang (1997),

$$g'(\theta_b) = \int_{-\infty}^{\infty} U'(\theta_b R_f + \theta_s R_s) R_f f(R_s, \mu_s, \sigma_s^2) dR_s = E \left[U'(\theta_b R_f + \theta_s \tilde{R}_s) R_f \right]. \quad (49)$$

We can likewise prove the F.O.C. for θ_s , and furthermore, the F.O.C.s for θ_b and θ_s in the CARA-Normal, CRRA-Lognormal, and most importantly, HARA-Lognormal cases.

Expected Utility and Mean-Variance Analysis

We have examined two approaches to the optimal portfolio choice problem, namely, Expected Utility and Mean-Variance Analysis.

Mean-Variance Analysis is intuitive hence popular both in practice and academia, while Expected Utility Framework is axiomatic hence deeply rooted in decision theory. How do the two relate to each other?

- Mean-Variance Analysis: Investors use variance to measure risk, and they trade off a higher expected return with a lower variance.
- Expected Utility Framework: Investors generally care about higher moments, like skewness and kurtosis, of returns.
- However, under certain assumptions on preferences and asset returns, both approaches lead to similar investment decisions.

Assumptions on Preferences and Asset Returns

In the one-period setup, we can approximate the utility function as

$$\begin{aligned} U(W_1) &= U(E(\widetilde{W}_1)) + U'(E(\widetilde{W}_1))(W_1 - E(\widetilde{W}_1)) \\ &\quad + \frac{1}{2}U''(E(\widetilde{W}_1))(W_1 - E(\widetilde{W}_1))^2 + \dots \\ &\quad + \frac{1}{n!}U^{(n)}(E(\widetilde{W}_1))(W_1 - E(\widetilde{W}_1))^n + \dots \end{aligned}$$

Assumptions on Preferences and Asset Returns

An investor's expected utility depends only on the mean and variance of her wealth/the portfolio return if

- she has the **quadratic utility** hence all the derivatives of order 3 or higher are equal to zero,
 - One undesirable property of quadratic utility: increasing absolute risk aversion w.r.t. wealth.
- or the return distribution is **normal** hence all the terms of order 3 or higher can be expressed by the mean and variance.
 - Note that two-parameter distributions include gamma, normal, lognormal, etc. The stable distributions allow a portfolio to have the same distribution as those of its underlying assets – the so called “additivity”. And the normal is the only stable distribution that has finite variance.

Recent Studies on Higher Moments*

The literature has accumulated empirical evidence on the effect of higher moments on asset pricing. For example,

— Skewness and Kurtosis:

- 1 Harvey, Campbell R., and Akhtar Siddique. Conditional skewness in asset pricing tests. *Journal of Finance* 55.3 (2000): 1263-1295.
- 2 Conrad, Jennifer, Robert F. Dittmar, and Eric Ghysels. Ex ante skewness and expected stock returns. *Journal of Finance* 68.1 (2013): 85-124.
- 3 Amaya, D., Christoffersen, P., Jacobs, K., and Vasquez, A. (2015). Does realized skewness predict the cross-section of equity returns? *Journal of Financial Economics*, 118(1), 135-167.

— Tail Risk:

- 1 Bollerslev, Tim, and Viktor Todorov. Tails, fears, and risk premia. *Journal of Finance* 66.6 (2011): 2165-2211.
- 2 Kelly, Bryan, and Hao Jiang. Tail risk and asset prices. *Review of Financial Studies* 27.10 (2014): 2841-2871.
- 3 Bollerslev, Tim, Viktor Todorov, and Lai Xu. Tail risk premia and return predictability. *Journal of Financial Economics* 118.1 (2015): 113-134.

Higher Order Derivatives of Utility Functions*

Write $[x, y]$ to denote a lottery with a 50-50 chance of receiving either outcome x or outcome y . We assume throughout that all individuals prefer more wealth to less. (The first derivative is positive, $U' > 0$.)

— Risk aversion:

Consider an individual with an initial wealth $W > 0$. Let $k_1 > 0$, $k_2 > 0$, and $W > k_1 + k_2$. All variables are defined so as to maintain a strictly positive total wealth.

Consider two lotteries, A: $[W - k_1, W - k_2]$ and B: $[W, W - k_1 - k_2]$.

An individual is risk averse ($U'' < 0$) if and only if lottery A is preferred to lottery B for all possible values of W , k_1 , and k_2 . Note that A and B have the same expected final wealth, but B has a higher variance than A. See Eeckhoudt, Louis and Harris Schlesinger, 2006, Putting Risk in Its Proper Place, *American Economic Review* 96, 280-289.

Prudence*

→ Prudence:

Consider an individual with an initial wealth $W > 0$. Let $k > 0$ and $\tilde{\omega}$ be any zero-mean random variable.

Consider two lotteries, A: $[W - k, W + \tilde{\omega}]$ and B: $[W, W - k + \tilde{\omega}]$.

Eeckhoudt and Schlesinger (2006) define prudence as a preference for lottery A over lottery B for every arbitrary W , k , and $\tilde{\omega}$. They also show how this lottery preference is equivalent to a convex marginal utility in expected-utility models, $U''' > 0$.

Temperance*

—o Temperance:

Consider an individual with an initial wealth $W > 0$. Let $\tilde{\omega}$ and $\tilde{\delta}$ be two independent zero-mean random variables.

Consider two lotteries, A: $[W + \tilde{\omega}, W + \tilde{\delta}]$ and B: $[W, W + \tilde{\omega} + \tilde{\delta}]$.

Eeckhoudt and Schlesinger (2006) define temperance as a preference for lottery A over lottery B for every arbitrary W , $\tilde{\omega}$, and $\tilde{\delta}$. They also show how this lottery preference is equivalent to a concave 2nd derivative in expected-utility models, $U'''' < 0$.¹²

¹²For even higher order derivatives, see: Deck, Cary, and Harris Schlesinger. Consistency of higher order risk preferences. *Econometrica* 82.5 (2014): 1913-1943.

Case II